What to expect in these two lectures

CV2090 Probability & Statistics for Civil Engineers · summary, lecture 2 of 2

Lecture 1

- The course map: all three branches at a glance
- 1. Uncertainty analysis
- 2. Learning from data
- For each branch: its parts and stories, which tool when, know this, traps the questions test

Lecture 2 (today)

- **3. Decision under uncertainty: its parts and stories, which tool when, know this, traps the questions test**
- **Ten ideas to keep**
- **Beyond the course: how the parts connect, practice and communication, ten questions to ask of any claim**
- **Where to go next**

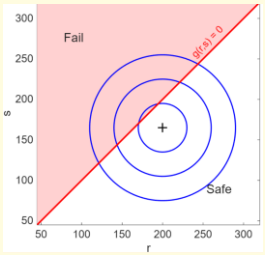
3. Decision under uncertainty (Reliability): four parts

Question: *Which one is safer?*

Reliability analysis

L-36–L-38

- Limit states: ultimate, serviceability, fatigue
- $g(R,S) = R - S$; $p_f = Pr[g < 0]$; reliability = $1 - p_f$
- p_f by conditioning on R or on S
- $\beta = \mu_Z/\sigma_Z$, $p_f \cong \Phi(-\beta)$; normal and lognormal cases

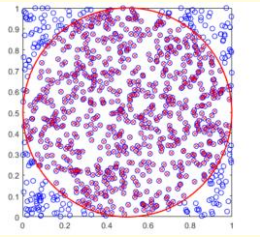


Story: structural column under D, L, W, E

Computational reliability

L-37

- "A MCS replaces an integral by an average"
- $\hat{p}_f$ = mean of the indicator q_i; unbiased
- c.o.v. $\cong \sqrt{(1/(N\ p_F))}$; sample size; stopping rule
- Area, volume, integrals: circle, 4-D ball



Stories: throwing points at a circle; Ulam's solitaire

Risk & decisions

partly covered · L-36

- Risk = expected loss; Probability × Consequence → Risk
- Target probabilities: 10^{-3} act now; 10^{-6} too rare
- "When everything is a priority, then nothing is!"
- Not covered: utility theory, Bayesian decision theory



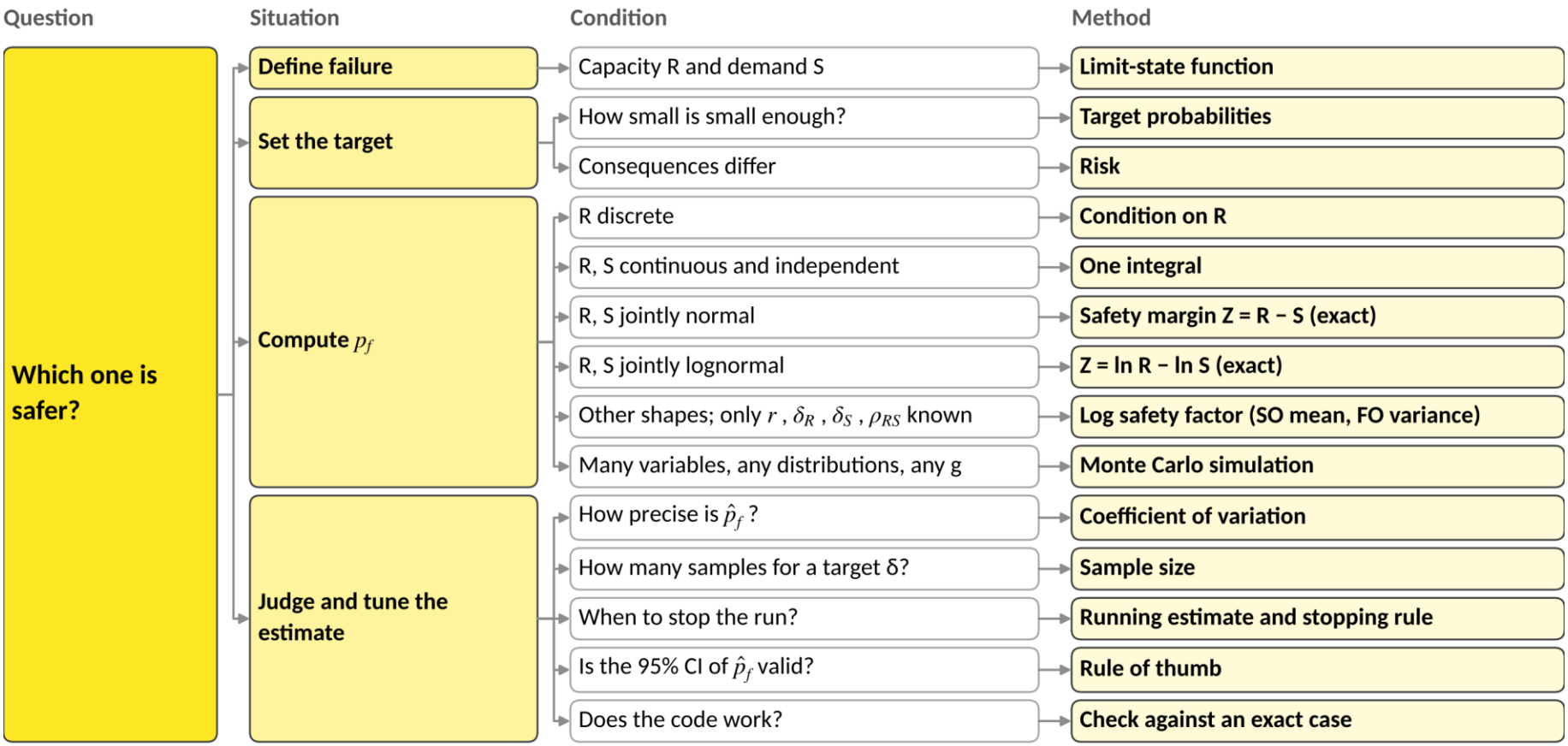
Stories: Challenger, Chernobyl, Fukushima, plane vs. car

Time-dependent reliability

not covered this term

- Listed in the Course Flow; no slides
- Related, taught in M1C4: Poisson model, return period, design life
- Syllabus topic 10 also lists Poisson processes and random walks

Which tool when? 3. Decision under uncertainty



Know this: 3. Decision under uncertainty (1 of 2)

Reliability analysis · risk and target probabilities

Know this

- The **limit-state function** $g(R,S) = R - S$ separates safe ($g > 0$) from failure ($g < 0$); $p_f = Pr[g < 0]$, reliability $= 1 - p_f$.
- Limit states: **ultimate** (loss of capacity), **serviceability** (deflection, vibration, cracking), **fatigue** (repeated loads).
- **Reliability index** $\beta = \mu_Z/\sigma_Z$; a larger β means a smaller $p_f \cong \Phi(-\beta)$.
- **Risk** is expected loss: Probability × Consequence → Risk.

Don't confuse

- *"Two bridges with the same FoS are equally safe."*
  → Same FoS = 1.67, different variability, different safety.
- *"The bridge is safe."*
  → "Given what we know today, the bridge has acceptably small probability of failure."
- *"Prepare equally for every 'what if'."*
  → "When everything is a priority, then nothing is!" (Challenger)
- *"β and p_f are the same number."*
  → $p_f = \Phi(-\beta)$. Case A: $\beta = 2.186$ gives $p_f = 0.01439$.

Know this: 3. Decision under uncertainty (2 of 2)

Exact and approximate p_f · Monte Carlo simulation

Know this

- $p_f = \Phi(-\beta)$ is **exact** for jointly normal R, S ($Z = R - S$) and jointly lognormal R, S ($Z = \ln R - \ln S$); the log safety factor is an approximation.
- "A MCS replaces an integral by an average"; for p_f, average the indicator $I(g(x) < 0)$.
- $\hat{p}_f$ is unbiased with $\mathrm{Var} = p_F(1 - p_F)/N$, so its c.o.v. $\cong \sqrt{(1/(N\, p_F))}$.
- Always report $\hat{p}_F(N)$, $\hat{\delta}(N)$ and N; stop when $\hat{\delta} \leq \delta_0$ or $N \geq N_0$.

Don't confuse

- *"Every β formula works for any distribution."*
  → Each row assumes a distribution shape. "Monte Carlo needs neither."
- *"A fixed N is enough for any p_f"*
  → Small p_F needs large N: the c.o.v. falls only as $1/\sqrt{N}$.
- *"The 95% CI of $\hat{p}_f$ is always valid."*
  → Only once $N\hat{p}$ and $N(1 - \hat{p})$ are both above about 10.
- *"If we can simulate physics, simulate everything."*
  → Combinations grow exponentially: 10 parameters × 10 values = 10^{10}.

Traps the questions test: 3. Decision under uncertainty

Failure probability

"Pr(roof collapse) = $f_S(5)$ = 24/125 = 0.192."

→ A density is not a probability: Pr(S > 5) = $\int_5^6$ 24/s³ ds ≈ 0.147. (Quiz-1 Q8)

Risk from rare states

"Severe surges are only 10% of seasons, so they add little to overtopping risk."

→ Pr(O) = 0.012 + 0.045 + 0.065 = 0.122; the severe state gives 53% of it. (Tut2 Q1)

Design for extremes

"Design the seawall for the observed maximum of 5.7 m with a margin."

→ The Gumbel model gives a mean annual maximum of 7.6 m and y_{1000} ≈ 15 m. (Tut5 Q2)

Design life

"Pr(at least one exceedance in 50 years) = 50 × annual p."

→ Use $1 - (1 - p)^{50}$: 2000 m³/s has p ≈ 0.032 (T ≈ 31 yr), so ≈ 0.80 in 50 years. (Quiz-2 Q7; Tut5 Q2e)

No tutorial or quiz question covers limit states, β or Monte Carlo (taught after Tutorial-8).

Course Feedback

Procedure for TCF submission in mobile application

i. Download and install **TCF app** from Google Play store (Android Devices)/ by searching "IITM TCF" in the App store (iOS devices).

ii. New user can "Register" with Roll No. Your password will be sent to your IITM smail account.

iii. On successful login, click on a course to enter your feedback. If the course is taught by more than one faculty, all the faculty members will be listed. Select the teacher who taught you the course.

iv. The first question in the questionnaire is whether you are willing to give feedback.

   a. If no, then give appropriate reason for the same.

   b. If yes, then go through the questionnaire and give appropriate feedback.

v. The TCF system includes the following:

   - Form- A: For theory courses with class strength 30 and above
   - Form- B: For Laboratory courses
   - Form-C: For theory courses with class strength less than 30

vi. In the final screen, click on "Submit" button to complete your feedback.

Ten ideas to keep

1. **Uncertainty comes from incomplete knowledge, not failed physics, and engineers must decide anyway.**

   Steel ball dropped 100 times

2. **A probability needs a stated model; change the model and the answer changes.**

   Bertrand paradox

3. **Probability, not plausibility: intuition often misleads.**

   Birthdays, Who is Reyansh?

4. **Evidence updates belief: posterior $\propto$ likelihood $\times$ prior.**

   O. J. Simpson, Covid test

5. **Summaries are not the data: equal μ, σ^2 and ρ can hide different patterns. Plot the data.**

   Datasaurus Dozen

6. **Dependence matters: joint behavior, not each variable alone, decides outcomes and the spread of sums.**

   Two mutual funds

7. **Sums tend to Normal, products to Lognormal, and maxima to Gumbel, Fréchet or Weibull.**

   Heights, wealth, floods

8. **Design for the extreme not yet observed: the record is not an upper bound.**

   Fukushima, 1953 North Sea flood

9. **Learn from data with stated uncertainty: an interval for a number, a test for a claim, a band for a prediction.**

   Darts, justice system, 25 storms

10. **Safety is a probability: replace "the bridge is safe" with an acceptably small p_f, and simulate when formulas fail.**

    Two bridges, Challenger

Beyond the course: How the parts connect

	Probability	Statistics	Decision
Direction	Model → data: if the model holds, what will we see?	Data → model: given what we saw, what is the model?	Model + data + consequences → action
Typical question	How likely is a 100-year flood within a 50-year design life?	What is the mean concrete strength, and how sure are we?	Raise the levee, or accept the risk?
Main tools	Distributions, conditioning, simulation	Estimates, intervals, tests, regression	Expected loss, p_f, target probabilities
Output	Probabilities of outcomes	Estimates with their uncertainty	An action and the risk it leaves
Typical failure	Wrong model: ignored dependence, tails too light	Biased sample, overfitting, many tests	Ignoring consequences; false precision
Feeds into	Supplies statistics with models	Supplies decisions with calibrated inputs	Tells the others what precision is worth paying for

Decisions create new data (inspections, monitoring), which update the models.

Practice and communication

In practice

- Plot the data first; check units, outliers and missing values
- Test a method on a case with a known answer before trusting it
- Check the result on data not used to build the model
- Ask which assumption changes the answer most (sensitivity)
- Keep data, code and random seeds so others can repeat the work

In communication

- Give every number its uncertainty and its basis (interval, sample size, method)
- State the time frame and the unit: per year, per lifetime, per person
- Natural frequencies are easier to grasp: “3 in 1,000 per year” rather than “0.003”
- Keep apart the probability of an event, its consequence, and the decision
- Say what the analysis does not cover

Ten questions to ask of any claim

1. What exactly is claimed, and compared with what?
2. Where did the data come from, and who was left out?
3. How large is the sample, and how different would another sample be?
4. Is this Pr(A | B), or Pr(B | A)?
5. What is the base rate?
6. Is the effect large enough to matter, not only “significant”?
7. Could chance, confounding or selection explain it?
8. How many things were tried before this one was reported?
9. What does the model assume, and what if that is wrong?
10. What would change the conclusion, and what does being wrong cost?

Where to go next

Topics that build on the course

1. Uncertainty analysis

- Copulas and tail dependence
- Non-stationary and climate extremes

2. Learning from data

- Goodness-of-fit tests and model selection (AIC)
- Multiple regression and Kriging

3. Decision under uncertainty

- System reliability
- Time-dependent reliability and risk-informed decisions

Books

Syllabus textbooks

- **Benjamin & Cornell** (1970), *Probability, Statistics and Decisions for Civil Engineers*: all three branches, for civil engineers
- **Ang & Tang** (Vol. I 1975, Vol. II 1984), *Probability Concepts in Engineering Planning and Design*: I: basic principles; II: decision, risk and reliability
- **Papoulis** (3rd ed., 1991), *Probability, Random Variables and Stochastic Processes*: the mathematics, through stochastic processes
- **Nowak & Collins** (2nd ed., 2012), *Reliability of Structures*: structural reliability beyond Module 3
- **Der Kiureghian** (2022), *Structural and System Reliability*: reliability methods, including systems

Syllabus popular reading (not exam material)

- **Kahneman** (2011), *Thinking, Fast and Slow*: how intuition misjudges chance (selected chapters)
- **Taleb** (2nd ed., 2010), *The Black Swan*: rare, high-impact events
- **Silver** (2012), *The Signal and the Noise*: why many predictions fail, and some don't

Thank you

CV2090 Probability & Statistics for Civil Engineers, Jul–Nov 2026

> *“When everything is a priority then nothing is.”*

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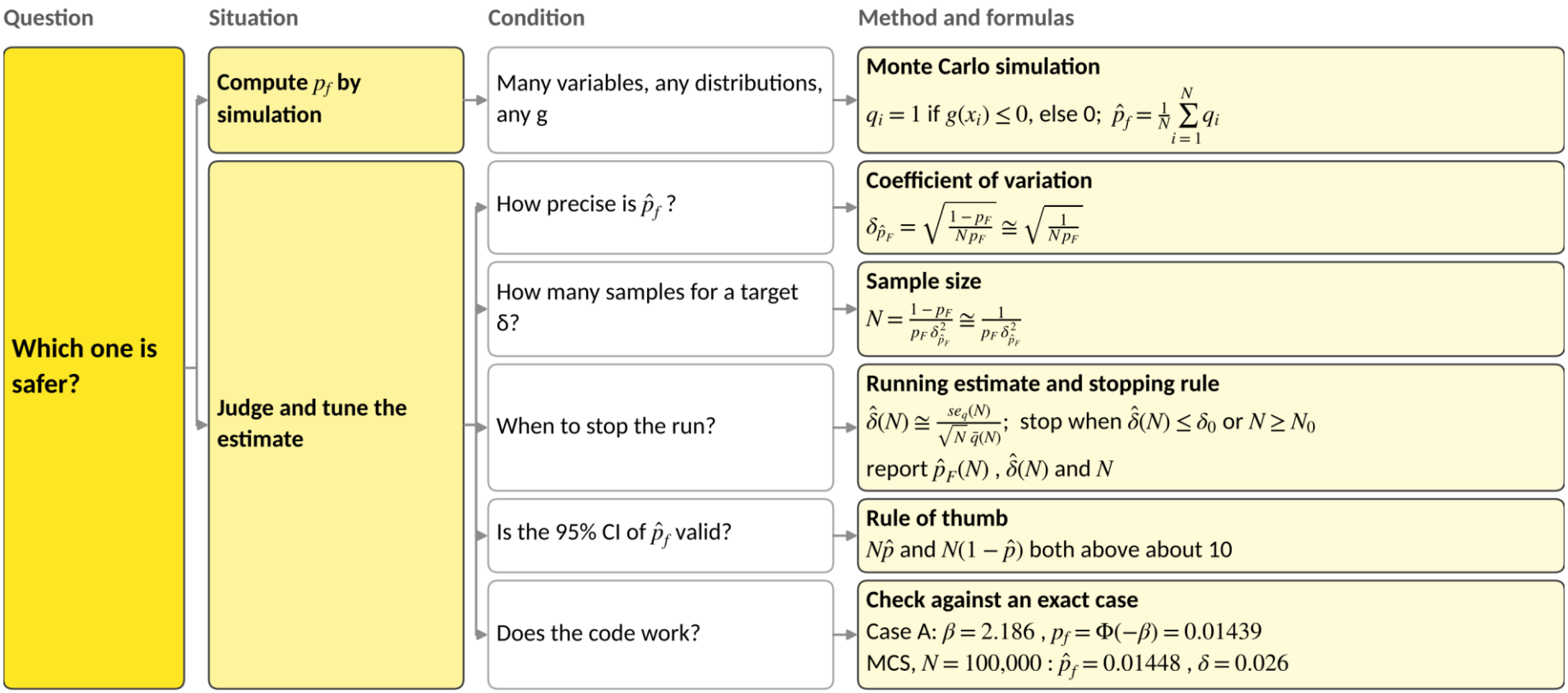
Space Shuttle Challenger disaster (Jan 28, 1986)

Thank you!

With formulas: 3. Decision under uncertainty (1 of 2)

Question	Situation	Condition	Method and formulas
Which one is safer?	**Define failure**	Capacity R and demand S	**Limit-state function** $g(R,S) = R - S$; reliability $= 1 - p_f$ $p_f = \Pr[g < 0] = \iint_{r \le s} f_{RS}(r,s)\,dr\,ds$
	Set the target	How small is small enough?	**Target probabilities** death/person/year: $> 10^{-3}$ act now; $< 10^{-6}$ little can be done building collapse $\sim 10^{-4}$ (lifetime); NPP core meltdown $\le 10^{-5}$ /year
		Consequences differ	**Risk** risk = expected loss; Probability × Consequence → Risk
	Compute p_f	R discrete	**Condition on R** $p_f = \sum_i \Pr(R = r_i)\,\Pr(R < S \mid R = r_i)$
		R, S continuous and independent	**One integral** $p_f = \int_0^\infty F_R(s) f_S(s)\,ds = \int_0^\infty [1 - F_S(r)] f_R(r)\,dr$
		R, S jointly normal	**Safety margin Z = R − S (exact)** $\beta = \frac{\mu_R - \mu_S}{\sqrt{\sigma_R^2 - 2\rho_{RS}\sigma_R\sigma_S + \sigma_S^2}}$, $p_f = \Phi(-\beta)$
		R, S jointly lognormal	**Z = ln R − ln S (exact)** $\mu_Z = \ln\left(r\sqrt{\frac{1+\delta_S^2}{1+\delta_R^2}}\right)$, $\sigma_Z^2 = \ln\frac{(1+\delta_R^2)(1+\delta_S^2)}{(1+\rho_{RS}\delta_R\delta_S)^2}$, $\beta = \frac{\mu_Z}{\sigma_Z}$
		Other shapes; only r , δ_R , δ_S , ρ_{RS} known	**Log safety factor (SO mean, FO variance)** $\beta \cong \frac{\ln r - \frac{1}{2}(\delta_R^2 - \delta_S^2)}{\sqrt{\delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2}}$, $r = \mu_R/\mu_S$

With formulas: 3. Decision under uncertainty (2 of 2)



What the subject is about

Probability describes what we do not know. Statistics learns about it from data. Engineering decides anyway, and says how sure it is.

Probability

- From a model to what we should expect to see
- *Given a wind model, how often will a tower see a design gust?*

Statistics

- From data back to the model, with stated uncertainty
- *From a set of cube tests, what is the mean strength, and how sure are we?*

Decision

- From models, data and consequences to an action
- *Is this bridge's failure probability small enough, or should we strengthen it?*

Experts disagree: Is a probability a long-run frequency (frequentist view) or a degree of belief updated by evidence (Bayesian view)? Engineers use both: frequencies for repeatable loads, degrees of belief for one-off events such as the rupture of a particular fault.

Habits of thinking

Ask “compared with what?”

Every rate needs a baseline: base rates, control groups, the same period last year.

Think in distributions

Report the spread and the tails, not only the average. A mean strength alone says little about failures.

State the model

Write down the assumptions (independence, a stable climate, a distribution shape) and ask what breaks if one is wrong.

Separate variability from ignorance

Natural scatter does not shrink with more data; lack of knowledge does.

Look at the data and the tails

Plot before you summarize. Rare, large events drive many failures.

Doubt causal stories

Correlation is not causation. Ice-cream sales and drownings both rise in summer.